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Grid Forming Fast Frequency Response for PMSG-based Wind Turbines

Xue Lyu, *Member, IEEE,* Dominic Groß, *Member, IEEE*

***Abstract*—Electric power systems are undergoing a rapid transition from fuel-based generation using synchronous generators to renewable generation interfaced by power electronics. In this context, a key challenge is using renewable generation with limited controllability to contribute to power system stability. In this work, we investigate grid-forming control of permanent magnet synchronous generator (PMSG) wind turbines. The proposed control and curtailment strategy supports the entire spectrum of standard functions of grid-following (e.g. maximum power point tracking (MPPT) and grid-forming control (e.g., primary frequency control) without explicit mode switching. A detailed case study is used to compare the performance of the proposed control at operating points corresponding to MPPT and frequency control with standard grid-forming and grid-following controls. The results demonstrate that (i) the proposed energy balancing grid forming control is self-synchronizing in MPPT mode, (ii) the (limited) energy storage and controllability of wind turbines can be adequately utilized to provide grid support, and (ii) the proposed control exhibits good performance under variable wind speeds and does not result in a significant increase of wind turbine fatigue loads.**

***Index Terms*—Wind turbine, grid-following control, grid-forming control, frequency regulation.**

I. Introduction

ELECTRIC power systems are undergoing an unprecedented transition towards sustainable zero-carbon power systems relying on massive integration of renewable energy resources and power electronics. Jeopardizing system stability has been identified as one of the main concerns as the share of renewables and power electronics will continue to grow significantly in the foreseeable future [1], [2]. The dynamics of conventional multi-machine power systems are pre-dominantly governed by the inertia, turbine, and control (e.g., governor and power system stabilizer) of synchronous generators (SGs) [1]–[4]. Thus, replacing SGs with renewable generation interfaced by power electronics affects the very core of power system operation, control, and stability analysis. Today, renewable resources typically use so-called *grid-following* control (GFL), that crucially relies on the assumption that the ac voltage (i.e., magnitude and frequency) at point of common coupling (PCC) is stable irrespective of the converter current injection. While GFL power converters can provide typical ancillary services [5]–[8] they typically rely on a phase-locked loop (PLL) that may jeopardize system stability [9], [10]. In this setting, stability and reliability heavily depends on the grid-support functions of the remaining SGs [11] as well as the grid coupling strength and can rapidly deteriorate due to the large-scale integration of GFL-interfaced resources [1], [3], [4].

As SGs are phased out due to increased deployment of converter interfaced renewable resources, it is reasonable to expect renewable resources to replace SG grid support functions and contribute to system stability. To this end, grid-supporting control requires renewable sources to provide grid-support instead of operating at their maximum power point (MPP). In particular, existing works highlight the significant potential of wind turbines (WTs) to contribute to frequency regulation, voltage regulation, and fault ride-through [5]–[8], [12]. Because frequency stability is one of the most severe problem resulting from large scale integration of converter interfaced power sources [13] we exclusively focus on frequency support in this work. Conceptually, the active power reference of PLL-based GFL control can be modified according to the PLL estimate of the rate of change of frequency (RoCoF) and frequency deviation to provide an inertia response and primary control. However, the above-mentioned approaches use PLL-based GFL control and inherit stability and performance limitations of PLLs [9], [10].

To address these challenges, *grid-forming* (GFM) converters are envisioned to be the cornerstone of future power systems [3], [14]–[17]. In contrast to GFL converters, GFM converters impose a well-defined and stable ac voltage waveform (i.e., magnitude and phase angle) at the PCC, autonomously synchronize, and vary their frequency and voltage magnitude to control to provide grid-support [3], [14]. The most prevalent GFM control is so-called droop control [14], [18] and virtual synchronous machine control [15], [19] that linearly trade off active power setpoint and frequency deviations, and reactive power and voltage magnitude deviations.

Comparisons demonstrate that grid-forming converters result in improved frequency regulation [20], system damping [21], and stability margins [22] relative to GFL control. Moreover, GFM WTs can successfully respond to frequency deviation events [23]. However, prevalent GFM converters are prone to instabilities if their dc voltage is not tightly controlled by their dc power source [16]. This requirement poses a challenge for WTs, e.g., GFM control can destabilize the system if the WT reaches its MPP instead of falling back to MPPT. Overall, the features of GFM and GFL control (i.e., stabilizing ac and dc terminals) are complementary and both GFM and GFL are needed to operate emerging power systems [24], [25]. At the same time, interoperability of GFL and GFM

X. Lyu is with Pacific Northwest National Laboratory, Richland, USA (e-mail: xuelu111@gmail.com)

D. Groß is with the Department of Electrical and Computer Engineering, University of Wisconsin-Madison, Madison, USA (e-mail: dominic.gross@wisc.edu)

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control is a concern [22], [26].

Instead of active power, the dc voltage can be used as a control signal to emulate the response of SGs [27]. This approach simultaneously stabilizes the ac and dc voltage while ensuring power balancing between the ac and dc terminal and can be used to replace both GFL and GFM control [24], [25]. While, the underlying idea has been applied to static synchronous compensators [28], VSC-HVDC systems [25], [29] and WTs operating at their MPP [30], it can also be applied to WTs to provide GFM grid-support.

In the context of GFL control, WTs with grid-support functions typically utilize energy storage systems, the residual kinetic energy of the WT rotor, and blade pitch control [31]–[33]. To the best of the authors' knowledge, the question of how appropriately leverage the degrees of freedom and inherent energy storage of WTs for GFM control has not received significant attention.

Therefore, in this work we develop a GFM control architecture for PMSG WTs that combines dual-port GFM control to imposes the ac voltage at the PCC with a suitable curtailment and machine/WT control strategy to fully leverage the WT grid-support capabilities across a wide range of operating points. Moreover, we demonstrate that the proposed control supports the entire spectrum of standard functions of GFL (e.g., MPPT) and GFM (e.g., primary frequency control and inertia) controls without switching controls.

We use high-fidelity simulation to compare the dynamic performance of different converter control strategies (i.e., standard GFM and GFL, energy-balancing with and without grid supporting) and study their properties when applied to PMSG WTs. The main contribution of this work can be summarized as follows:

1) A novel GFM control is proposed that accounts for WT dynamics and WT limits, subsumes standard GFL and GFM functions, and maintains stable operation when the WT reaches its power limits without relying on additional energy storage or grid frequency estimates (i.e., a PLL).
2) A curtailment strategy is proposed to fully leverage the WT frequency control capabilities.
3) The proposed GFM control is agnostic to the WT operating point and seamlessly (i.e., without switching controls) supports the entire range from MPPT to frequency support by adjusting a WT curtailment factor.
4) The proposed control improves small-signal stability [34] and dc voltage damping by means of a derivative dc voltage-frequency droop (i.e., dc voltage to angle droop).

This manuscript is structured as follows. Sec. II introduces the problem setup and basic terminology. In Sec. III, we review GFL MPPT control of PMSG WTs. The proposed curtailment strategy and controls are presented in Sec. IV. The proposed control is illustrated and compared to standard controls using a case study in Sec. V. Finally, Sec. VI provides the conclusions and discusses future research directions.

II. Problem Setup

A. Terminology

Before describing the problem setup in more detail, we emphasize that there is no precise and universally agreed upon definition of GFM and GFL across different research communities. In parts of the literature a dc/ac converter is called grid-forming if it creates an ac voltage with constant amplitude and frequency [14]. The same definition is also widely used in the context of offshore converters in offshore wind. Other definitions encountered in the power systems literature (e.g., [4]) hinge on the presence of virtual inertia, the ability to operate an islanded ac grid, the ability to suppress frequency oscillations [35], or the absence of a PLL [30]. However, several controls exist that are arguably GFM but use an auxiliary PLL [15]. Finally, in e.g., in HVDC and offshore wind, grid-forming capabilities may also refer to creating a stable dc voltage. Attempting to resolve the conceptual differences between these definitions is beyond the scope of this manuscript. Instead, to broadly categorize converter controls, we use the following definition adapted from [25].

Definition 1 (Grid-forming and grid-following control) *For each converter terminal (i.e., ac and dc), we refer to a converter as GFM if it imposes a stable voltage at the converter terminal (i.e., ac-GFM or dc-GFM), and GFL if it has to rely on another device to impose a stable voltage at the converter terminal (i.e., ac-GFL or dc-GFL).*

B. PMSG WT control architectures

In this work, we propose a novel approach to ac-GFM/dc-GFM control of PMSG-WTs that leverages (i) ideas from machine emulation control [27] and dual-port GFM control [24], [25] to control the grid-side converter (GSC) of a PMSG WT and, (ii) develops suitable curtailment strategies and machine-side converter (MSC) controls to realize standard grid-support functions (i.e., inertia response, primary frequency control). The remainder of this section discusses the main salient features of the proposed control architecture relative to standard controls.

We first consider the widely used PLL-based ac-GFL control with MPPT shown in Fig. 1a that does not provide frequency support. In this case, the MSC is controlled to extract the maximum available power from the wind, i.e., its power injection into the dc-link is largely uncontrolled. The GSC utilizes PLL-based ac-GFL/dc-GFM control (i.e., assuming stable ac voltage at the PCC) to tightly control the dc-link voltage through its ac active power injection. Instead, energy-balancing dual-port GFM (i.e., ac-GFM/dc-GFM) control [24], [25] can be used to realize MPPT as shown in Fig. 1b. Notably, similar to [30], energy-balancing GFM imposes a well-defined and stable ac voltage waveform at the GSC terminal without using a PLL (i.e., does not have to rely on a stable ac voltage).

To provide frequency support functions, the controls in Fig. 2 combine a curtailment strategy with ac-GFM controls. Standard ac-GFM/dc-GFL GSC control [15], [18], [36] that requires a stable dc voltage is combined with proportional-integral (PI) control of the dc voltage through the MSC in Fig. 2a. We emphasize that this control architecture cannot operate at the MPP because, at this operating point, the dc voltage can only be stabilized through the GSC. In other words, if the WT has no reserves (i.e., reaches it MPP) the dc-link voltage is uncontrolled and may become

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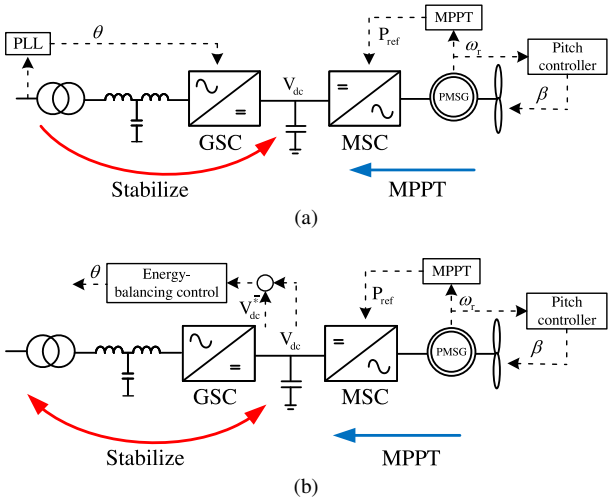


Fig. 1. Maximum power point tracking for PMSG WT using standard ac-GFL/dc-GFM GSC control (a) and dual-port GFM GSC control (b).

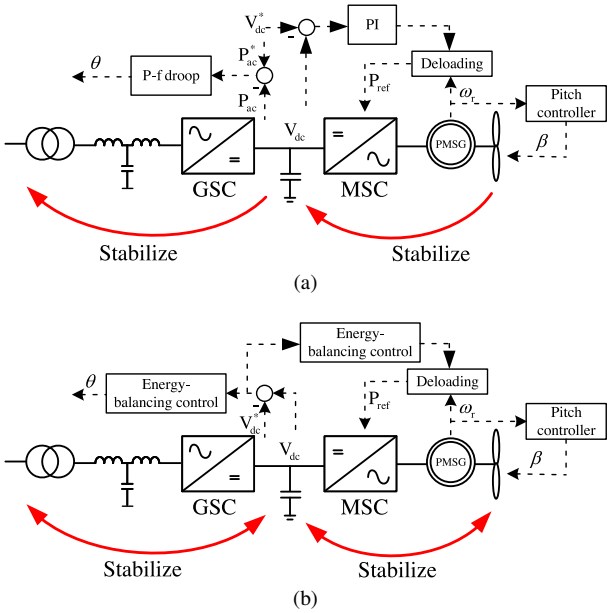


Fig. 2. Grid supporting control of PMSG WT using standard ac-GFM/dc-GFL GSC control (a) and dual-port GFM GSC control (b).

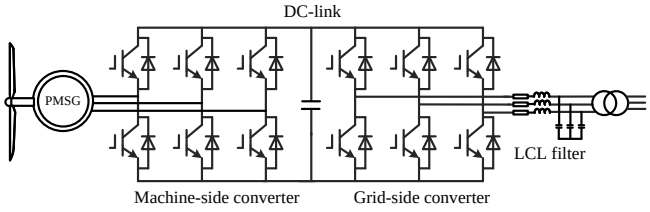


Fig. 3. PMSG-WT system with two two-level voltage source converters.

unstable. Finally, dual-port GFM control is combined with a proportional dc voltage through the MSC in Fig. 2b. In this control architecture, power imbalances propagate from the grid to the dc-link and the MSC implicitly results in an inertia support and primary frequency control at the GSC.

The conceptual disadvantage of standard ac-GFM/dc-GFL GSC control is that it decouples the WT frequency response (as imposed by the GSC control) from the capabilities of the power generation (i.e., the WT). In contrast, the dual-port GFM control in Fig. 1b and Fig. 2b propagates any grid power imbalances to the power generation (i.e., the MSC and WT) which responds according to its capabilities. The main contribution of this work is a curtailment strategy and WT control tailored to dual-port GFM GSC control that fully leverages the WT capabilities to provide a frequency response while not overloading the WT or collapsing the dc-link voltage. A key advantage of the dual-port PMSG WT control in Fig. 2b is that it supports the entire range from MPPT to grid-support functions by changing the WT curtailment without any control switching. In contrast, the standard controls in Fig. 1a and Fig. 2a required online control switching depending on the services provided by the WT at the current time. Moreover, combining standard ac-GFM and ac-GFL controls on the same system can result in significant stability challenges [22], [26].

In the remainder, we present the details of each control architecture, and illustrate and compares their performance.

III. Grid-following MPPT Control of PMSG WTs

We consider a type 4 WT that contains a WT, PMSG, back-to-back connected voltage source converters (VSC) (i.e., machine-side and grid-side), dc-link capacitor, grid-side LCL filter and transformer. The components of the type 4 PMSG-WT are depicted in Fig. 3. In this section, we present the model of PMSG WT and review standard GFL MPPT control.

A. PMSG wind turbine model

The one-mass model of drive train is used in this work and the rotor dynamics of the wind turbine is governed by the swing equation

$$J\omega_{\mathrm{r}}\tfrac{\mathrm{d}}{\mathrm{d}t}\omega_r = P_{\mathrm{wt}} - \omega_{\mathrm{r}}T_{\mathrm{e}}. \tag{1}$$

where ω_{r} is rotor speed, and J is the combined PMSG and WT inertia, T_{e} is the electromagnetic torque of PMSG. Moreover, P_{wt} is the mechanical power captured by WT that can be expressed as

$$P_{\mathrm{wt}}(\beta,\omega_{\mathrm{r}},v_{\mathrm{w}}) = \tfrac{1}{2}\rho\pi R^2 C_p(\lambda,\beta)v_{\mathrm{w}}^3, \tag{2}$$

where ρ is the density of air, R is the blade length, v_{w} denotes wind speed, and $C_p(\lambda,\beta)$ is the power conversion coefficient determined by tip speed ratio

$$\lambda := \frac{\omega_r R}{v_{\mathrm{w}}}, \tag{3}$$

and the blade pitch angle β. The maximum power coefficient $C_{p,\max} := C_p(\lambda_{\mathrm{opt}},0)$ is attained at the optimal tip speed ratio λ_{opt} and $\beta = 0$ (see Fig. 4). Considering (3) and (2), the power P_{wt} can, within limits, be adjusted through the rotor speed ω_{r} and pitch angle β. Finally, detailed models of the PMSG stator current dynamics and the electromagnetic torque can be found in [38, Sec. II].

Assuming that the converter losses are negligible, the dc-link capacitor charge dynamics are given by

$$C_{\mathrm{dc}}v_{\mathrm{dc}}\tfrac{\mathrm{d}}{\mathrm{d}t}v_{\mathrm{dc}} = P_{\mathrm{msc}} - P_{\mathrm{gsc}}. \tag{4}$$

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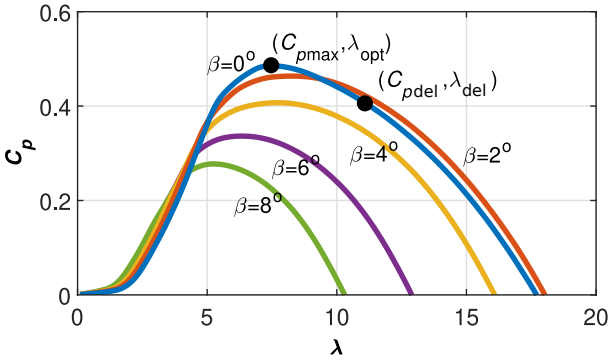


Fig. 4. Power coefficient C_p of a 5 MW wind turbine [37] as function of the pitch angle β and tip speed ratio λ. $(C_{p,\max}, \lambda_{\text{opt}})$ denote the MPP, and $(C_{p,\text{del}}, \lambda_{\text{del}})$ denote a curtailed operating point with increased rotor speed.

where C_{dc} is the dc-link capacitance, P_{msc} is the active power from MSC and P_{gsc} is the active power flowing into the grid. Notably, v_{dc} can be controlled either through P_{msc}, P_{gsc}, or both. Moreover, for the two-level voltage source converter in Fig. 3, the dc voltage v_{dc} needs to be larger than twice the ac voltage magnitude and smaller than the breakdown voltage of the semiconductor switches and dc-link capacitor. Typically, the dc voltage control enforces significantly tighter limits on the dc voltage than the hardware limits.

B. AC-GFL/DC-GFM MPPT Control of PMSG WTs

In conventional GFL MPPT control, the machine-side converter (MSC) controls the active power of PMSG to track the WTs optimal tip speed ratio λ_{opt}. In particular, the active power reference is generated via the MPPT algorithm,

$$P_{\text{mpp}} = \begin{cases} \frac{1}{2}\rho\pi R^5 \frac{C_{p\,\max}}{\lambda_{\text{opt}}^3}\omega_{\text{r}}^3 & \text{if } v_{\text{w}} < v_{\text{w}}^{\text{nom}}, \\ P_{\text{nom}} & \text{if } v_{\text{w}} \geq v_{\text{w}}^{\text{nom}}, \end{cases} \tag{5}$$

where $v_{\text{w}}^{\text{nom}}$ denotes the wind speed at which the MPP $P_{\text{mpp}}(v_{\text{w}}^{\text{nom}})$ coincides with the WT's nominal power, i.e., $P_{\text{mpp}}(v_{\text{w}}^{\text{nom}}) = P_{\text{nom}}$. Moreover, if $v_{\text{w}} < v_{\text{w}}^{\text{nom}}$, then $\beta = 0$, and, for $v_{\text{w}} \geq v_{\text{w}}^{\text{nom}}$ the pitch angle β is controlled via PI control to limit the rotor speed to its maximum. Finally, a vector current control in a dq-frame aligned with the PMSG angle is used to control the PMSG active power [39].

The MSC control crucially hinges on the assumption that the dc-link voltage v_{dc} is controlled to its nominal value by the GSC. The standard MPPT control uses a GSC vector current control in a dq-frame (aligned with a PLL grid voltage phase angle estimate) to control the dc-link voltage through the GSC current injection [14].

As discussed before, PLL-based ac-GFL control crucially hinges on the assumption that the ac voltage frequency and magnitude at the PCC are stabilized by other sources in the system, i.e., that a stable grid voltage with stiff frequency is formed by other devices. If this is not the case, the PLL-based GSC control will generally be unable to maintain stable operation [9], [10]. In principle, the PLL-based ac-GFL control can be combined with a suitable curtailment strategy to provide primary frequency control and inertia support. Specifically, the PLL can be used to obtain an estimate of the grid frequency and rate of change of frequency (RoCoF) and adjust the power generation of the WT and power injection of the GSC accordingly (see e.g., [5]–[7]). However, compared to ac-GFM control or synchronous generators, ac-GFL frequency support functions that rely on PLL frequency measurements typically result in frequency regulation with decreased performance, decreased system damping, and decreased stability margins (see e.g., [20]–[22]). Therefore, we will focus on ac-GFM grid-support functions in the remainder of this manuscript and use ac-GFL MPPT control as a baseline for comparisons.

IV. Grid-Forming Control of PMSG Wind Turbines

The main contribution of this work is a universal ac-GFM/dc-GFM control approach that supports both approximate MPPT and frequency regulation (e.g., fast frequency response). This section presents the curtailment strategy used as a basis for ac-GFM control with frequency support, discusses application of standard ac-GFM controls to PMSG-WTs, and presents our novel universal ac-GFM/dc-GFM control approach.

A. Curtailment strategy

In contrast to MPPT control, a power source with headroom for upward/downward regulation is required to provide frequency regulation services and grid-support. Specifically, to enable upward/downward regulation of the WT turbine power generation and provide an appropriate level of frequency regulation support, WTs typically need to operate away from the MPP. As shown in Fig. (4), the mechanical power P_{wt} captured from the wind depends on the rotor speed and pitch angle. Therefore, wind power curtailment can be achieved via adjusting the rotor speed and pitch angle.

The difference between these two approaches is that a considerable amount of kinetic energy can be stored in the form of kinetic energy by accelerating the rotor beyond the MPP when the available wind power is not fully passed on to the grid (i.e, $P_{\text{wt}} > \omega_{\text{r}} T_{\text{e}} = P_{\text{ref}} = P_{\text{ac}}$ in (1)). Depending on the wind speed, increasing the rotor speed ω_{r} beyond $\omega_{\text{r,opt}}$ can provide significant kinetic energy storage that can be used to provide grid support. In contrast, pitch control does not leverage the WTs inherent kinetic energy storage and can only provide downward regulation of the wind power generation. To find the optimal operation point of wind turbine with respect to the amount of kinetic energy stored in the rotating mass, the rotor speed control should have a high utilization priority.

In this work, we define the curtailment parameter $\eta \in \mathbb{R}_{[0,1]}$. Next, let $\bar{v}_{\text{w}}$ denote the wind speed above which $P_{\text{wt}}(0, \omega_{\text{rmax}}, v_{\text{w}})$ exceeds $\eta P_{\text{mpp}}(v_{\text{w}})$, i.e., pitch control needs to be used to limit the rotor speed. In other words, given η, $\bar{v}_{\text{w}}(\eta)$ is the (unique) solution of

$$\eta P_{\text{mpp}}(v_{\text{w}}) = P_{\text{wt}}(0, \omega_{\text{rmax}}, v_{\text{w}}). \tag{6}$$

Next, we propose to use the curtailed power reference

$$P_{\text{del}} = \begin{cases} \frac{1}{2}\rho\pi R^5 \frac{C_{p\text{del}}}{\lambda_{\text{del}}^3}\omega_{\text{r}}^3 & \text{if } v_{\text{w}} < \bar{v}_{\text{w}}(\eta), \\ \eta P_{\text{mpp}}(v_{\text{w}}) & \text{if } \bar{v}_{\text{w}}(\eta) < v_{\text{w}} \leq v_{\text{w}}^{\text{nom}}, \\ \eta P_{\text{nom}} & \text{if } v_{\text{w}} > v_{\text{w}}^{\text{nom}}, \end{cases} \tag{7}$$

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where $C_{p\text{del}} := \eta C_{p\max}$ and $\lambda_{\text{del}} \geq \lambda_{\text{opt}}$ denotes the tip speed ratio at which C_p equals $C_{p\text{del}}$ for zero pitch angle, i.e., $C_p(\lambda_{\text{del}}, 0) = \eta C_{p\max}$. In practice, $C_p(\lambda, \beta)$ is typically represented by a look-up table that can be used to compute λ_{del}. Moreover, we construct a look-up table that represents P_{del} as a function of the rotor speed ω_{r}, wind speed v_{w}, and the curtailment parameter η.

Using P_{del} as power reference for the MSC (i.e., $P_{\text{ref}} = P_{\text{del}}$), the rotor speed response for $v_{\text{w}} \leq \bar{v}_{\text{w}}(\eta)$ is similar to the MPPT control discussed in Sec. III, but converges to a tip speed and pitch angle with the pre-specified curtailment η. In contrast, if $v_{\text{w}} \geq \bar{v}_{\text{w}}(\eta)$, rotor accelerates to ω_{rmax} and the pitch angle β automatically increases to achieve the desired level of curtailment. Figure 5 illustrates the resulting look up table for P_{del} as a function of ω_{r} and v_{w} for 10% curtailment (i.e., $\eta = 0.1$) and a 5 MW wind turbine [37].

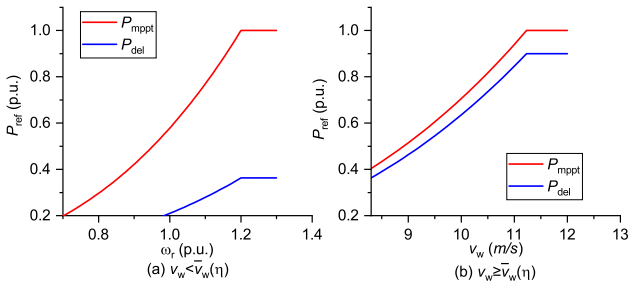


Fig. 5. The proposed curtailment strategy with 10% reserves (i.e., $\eta = 0.1$).

B. Grid-forming droop control for PMSG WTs

Conceptually, an ac-GFM control can be realized by combining dc-link voltage control through the MSC with ac-GFM/dc-GFL control [15], [18], [36] for the GSC. The most prevalent ac-GFM/dc-GFL control is so-called droop control

$$\omega_{\text{gsc}} = \omega_0 + \frac{K_p^{\text{ac}}}{T_{\text{ac}} s + 1}(P_{\text{ac}}^{\star} - P_{\text{ac}}), \tag{8}$$

with nominal system frequency ω_0 and active power setpoint $P_{\text{ac}}^{\star} = P_{\text{del}}$ set to the curtailed operating point (see Sec. IV-A). Moreover, K_p^{ac} is the active power droop coefficient and T_{ac} is a low-pass filter time constant. The GSC ac voltage phase angle θ is obtained by integrating ω_{gsc} (see Fig. 6).

We emphasize that most grid-forming controls have a very similar response when using realistic control gains (see [16] and [40, Sec. 3.4] for further details). For instance, (dispatchable) virtual oscillator control reduces to droop control during normal operation [41]. Similarly, droop control (8) is equal to a virtual synchronous machine with damping constant $D = 1/K_p^{\text{ac}}$ and inertia constant $H^{\text{gsc}} = T_{\text{ac}} D/2$ [19]. In other words, by suitable choice of T_{ac}, droop control (8) can also be interpreted as virtual synchronous machine (VSM) control with virtual inertia constant H^{gsc}. Therefore, we will focus on droop control and VSM control to highlight the key features of a wider class of grid-forming controls.

An active power limiting control (see, e.g., [21]) can be added as shown in Fig. 6 to limit the active power injection from the GSC. The limit $P_{\max}$ is typically set to (i) the rated power of the GSC in order to protect the GSC, (ii) or to P_{mpp} to avoid instability due to overloading the generation source interfaced by the GSC. Finally, the GSC ac voltage magnitude reference V_{ac} is given by the volt-var droop control

$$V_{\text{ac}} = V_{\text{ac}}^{\star} + \frac{K_q}{T_{\text{ac}} s + 1}(Q_{\text{ac}}^{\star} - Q_{\text{ac}}), \tag{9}$$

with voltage magnitude reference $V_{\text{ac}}^{\star}$, reactive power setpoint $Q_{\text{ac}}^{\star}$, and reactive power droop coefficient K_q. We emphasize that all ac-GFM controls discussed in this manuscript use volt-var droop control. However, for transmission connected plants frequency and active power and voltage and reactive power are commonly assumed to be decoupled during normal operation and hence we do not investigate the response of the system voltage or wind turbine reactive power injection in detail.

A significant drawback of standard ac-GFM/dc-GFL control is that it assumes a constant dc voltage v_{dc} [16]. In practice, this assumption requires that the dc voltage is stiffly regulated by a dc power source that can always provide the required power injection to stabilize the dc voltage. Conceptually, the

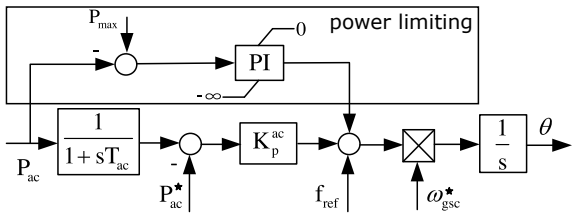


Fig. 6. Grid-forming GSC control through frequency-watt droop control with (optional) power limiting.

MSC can, within the operational limits of the WT, be used to stabilize the dc voltage v_{dc} by controlling P_{pmsg} as shown in Fig. 7. Specifically, we use proportional-integral (PI) dc voltage control

$$P'_{\text{ref}} = P_{\text{del}}(v_{\text{w}}) + \left(K_p + \frac{K_I}{s}\right)(v_{\text{dc}} - v_{\text{dc}}^{\star}) \tag{10}$$

with proportional gain K_p and integral gain K_I. However, in steady-state P_{pmsg} is limited by $P_{\text{mpp}}(v_{\text{w}})$ and the additional energy that can be provided by releasing kinetic energy from the rotor is limited by the requirement $\omega_{\text{r}} \geq \omega_{\text{r,opt}}(v_{\text{w}})$. Notably, if $\omega_{\text{r}} \leq \omega_{\text{r,opt}}(v_{\text{w}})$, the wind turbine is prone to instability because a drop in turbine speed results in a droop in power generation. To avoid this problem as well as excessive rotor deceleration that can damage the wind turbine, we introduce the dynamic limit

$$P_{\max}^{\text{dyn}} = P_{\text{mpp}}(v_{\text{w}}) + K_\omega(\omega_{\text{r}} - \omega_{\text{ropt}}(v_{\text{w}})) \tag{11}$$

with control gain K_ω for the power injection that decreases to $P_{\text{mpp}}(v_{\text{w}})$ as the rotor speed approaches the MPP speed $\omega_{\text{r,opt}}(v_{\text{w}})$. The outer MSC control with dynamic limiting (i.e., $P_{\text{mppt}} = \min\{P'_{\text{pmsg}}, P_{\max}^{\text{dyn}}\}$) is shown in Fig. 7.

Overall, this approach crucially relies on dc voltage control through the MSC. Therefore, it cannot support MPPT control and risks instability if the WT does not have sufficient reserves to stabilize the dc voltage (see e.g., [16]). This instability

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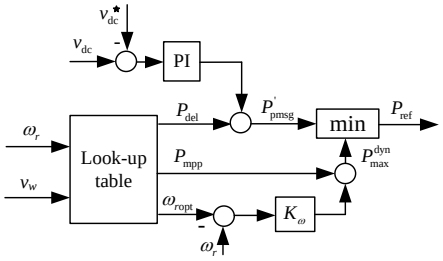


Fig. 7. DC voltage control through the MSC power reference.

can typically be avoided by setting $P_{\max} = P_{\text{mpp}}$ and always operating the WT with sufficient curtailment. However, in this case the kinetic energy stored in the WT rotor is not leveraged for inertia support and frequency control. Moreover, it should be noted that the additional limiter results in a switched control that is difficult to analyze and tune. Instead, the next subsection introduces a simpler control strategy that supports the entire spectrum from MPPT control to grid-support functions that fully leverage the kinetic energy stored in the WT rotor.

C. Energy-balancing dual-port GFM Control for PMSG WTs

To alleviate the issues observed in the previous subsection, we propose to leverage the concept of dc-voltage to frequency droop controls [24], [25], [27], [30], [42], [43]. In particular, energy-balancing control [25] for the GSC is combined with a suitable MSC control and the curtailment strategy discussed in Sec. IV-A.

Considering (4), the mismatch between wind power generation and electrical power injection leads to dc-link voltage variation. In particular, if the GSC injects more power into the grid than the MSC supplies, the dc-link voltage will collapse. This dynamic process is similar to frequency dynamics of a synchronous generator [27], [42], [43]. However, in the context of a PMSG WT, these controls emulate a synchronous generator with negligible inertia and damping. Instead, we propose to use the so-called energy-balancing dual-port GFM (i.e., ac-GFM/dc-GFM) control [25] that controls the dc voltage through the realizable proportional-derivative (PD) controller[1]

$$\omega_{\text{gsc}} = \omega_0 + \left(K_\theta + \frac{K_d^{\text{gsc}} s}{T_{\text{dc}} s + 1}\right)(v_{\text{dc}} - v_{\text{dc}}^{\star}), \qquad (12)$$

for the GSC ac voltage frequency with nominal system frequency ω_0, dc voltage setpoint $v_{\text{dc}}^{\star}$, proportional and derivative gains K_θ, K_d^{gsc}. Notably, the $v_{\text{dc}} - f$ droop with gain K_θ aims to stabilize the dc voltage and $\frac{\text{d}}{\text{d}t} v_{\text{dc}} - f$ droop induces additional dc voltage damping and improves small-signal stability [34]. Tuning guidelines for the control gains are discussed in Sec. IV-E. In contrast to VSM control, T_{dc} is a small low pass filter time constant used to suppress switching harmonics and the inertia response of (12) is solely determined by the curtailment and WT response.

[1] $\theta_{\text{gsc}} = \omega_0 t + (\frac{K_\theta}{s} + \frac{K_d^{\text{gsc}}}{T_{\text{dc}} s + 1})(v_{\text{dc}} - v_{\text{dc}}^{\star})$ can be used to avoid derivatives.

The GSC control (12) is combined with volt-var droop (9) and the overall control diagram is shown in Fig. 8. Notably, instead of relying on a well-defined ac voltage at the PCC for grid-following control, energy balancing (12) directly controls the GSC ac voltage frequency and magnitude and resembles an ac voltage source whose frequency varies with power imbalance and implicitly controls the dc voltage of the back-to-back converters through its ac terminal. In other words, if no

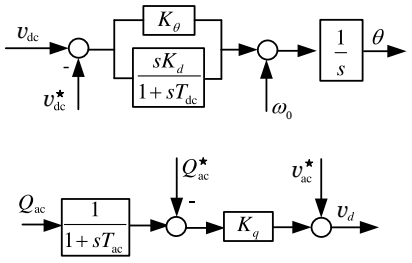


Fig. 8. Energy-balancing control is combined with volt-var droop to obtain references for the GSC ac voltage phase angle and voltage magnitude.

frequency regulation service is required, MPPT control can be used on the MSC. On the other hand, wind turbines can fully contribute to frequency regulation if the GSC control in Fig. 8 is combined with a suitable MSC control that implements the curtailment strategy discussed in Sec. IV-A.

In particular, we propose to control the dc voltage v_{dc} through the MSC power reference, i.e.,

$$P'_{\text{ref}} = P_{\text{del}}(v_{\text{w}}) + \frac{K_p^{\text{msc}} + K_d^{\text{msc}} s}{T_{\text{dc}} s + 1}(v_{\text{dc}}^{\star} - v_{\text{dc}}), \qquad (13)$$

where $P_{\text{del}}(v_{\text{w}})$ is the active power reference generated according to Fig. 5 and a PD dc voltage control with gains K_p^{msc} and K_d^{msc}. This approach may result in excessive power references that could damage the WT or result in unstable operation at tip speed ratios below the optimal tip speed ratio λ_{opt}. Therefore, an active power limit as given in (11) is used in MSC controller. Specifically, the dynamic limit $P_{\max}^{\text{dyn}}$ approaches $P_{\text{mpp}}(v_{\text{w}})$ as ω_{r} approaches $\omega_{\text{r,opt}}(v_{\text{w}})$, i.e., reduces the rotor deceleration to zero as ω_{r} approaches $\omega_{\text{r,opt}}(v_{\text{w}})$. The active power reference $P_{\text{ref}} = \min\{P'_{\text{ref}}, P_{\max}^{\text{dyn}}\}$ (see Fig. 9) is then tracked by the vector current control.

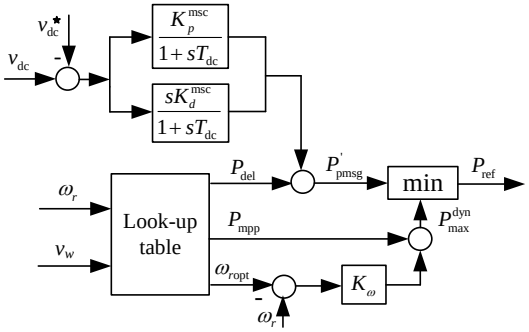


Fig. 9. DC voltage control through the MSC power reference.

Remark 1 (DFIG wind turbines) *Conceptually, energy-balancing control can be applied to DFIG wind turbines*

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by using the MSC rotor current control to implement the curtailment strategy in Sec. IV-A and stabilize the dc voltage. However, due to the limited converter rating of typical DFIG wind turbines, the rotor speed can only vary approximately $\pm 33\%$ around the synchronous speed [44]. This may limit the ability to provide the grid services (e.g., inertia support) discussed in this work to a narrow wind speed range.

D. Interpretation of energy-balancing control

In this section, we will discuss the physical synchronization mechanism of energy-balancing control that, depending on the operating point and MSC control, results in an inertia response and frequency regulation.

1) Synchronization mechanism: The dc voltage dynamics (4) reflect the active power balance between the GSC (i.e., demand) and MSC (i.e., power generation). Thus, using the GSC control (12), a load increase in the system results in a decrease of dc voltage, decrease of GSC frequency, and, ultimately, the GSC ac voltage phase angle converges to a phase angle configuration for which active power of the GSC (i.e., demand) and MSC (i.e., power generation) are balanced. This process mimics the self-synchronizing dynamics of SGs (see, e.g., [27] and [40, Sec. 2.2]. In particular, if the GSC control (12) is combined with MPPT control at the MSC, the GSC control will adjust the GSC ac voltage phase angle to pass the active power injection of the MSC to the ac system. We emphasize that the deviation of the dc voltage will be proportional to the system frequency deviation.

2) Inertia response: If the MSC tracks the power reference (13), a drop in system frequency (i.e., drop in dc voltage) results in a power injection from the MSC that is approximately proportional to the dc voltage deviation up to the limits imposed by and (11), i.e., the MSC power can exceed the MPP by the excess kinetic energy stored in the WT rotor. For wind speeds below the rated speed, the curtailment strategy proposed in Sec. IV-A prioritizes curtailment through rotor speed and significant kinetic energy can be stored in the rotor. In other words, for wind speeds below rated speed, energy-balancing control for PMSG WTs releases excess kinetic energy from the rotor to provide an inertia support to the system. In contrast, at above rated wind speed the optimal rotor speed $\omega_{r,\text{opt}} = \omega_{\max}$ and no inertia support can be provided.

3) Fast frequency response: In addition to the aforementioned inertia response, the curtailment strategy presented in Sec. IV-A and MSC power reference (13) result in a fast primary frequency control response. In particular, the $V_{\text{dc}} - P$ droop in (13) stabilizes the dc voltage and, implicitly, results in an increased power injection into the ac system that stabilizes the system frequency. Depending on the operating point, a drop in dc voltage results in a rotor speed and/or pitch angle decrease to capture more mechanical power from the wind. This increased power generation is passed to the system by the GSC and results in a fast primary frequency response that improves the rate of change of frequency (RoCoF), frequency nadir, and steady-state frequency after a load step. In cases when the system frequency deviation (i.e., dc voltage deviation) is large, the active power limit (11) ensures that the WT power injection does not exceed the MPP on longer time scales.

E. Tuning guidelines for energy-balancing control

To discuss the tuning of the GSC and MSC control gains, we first investigate the steady-state response of the WT to grid frequency deviation. Letting $s = 0$ in (12) and (13), the steady-state relationship between the GSC ac voltage frequency, wind turbine active power, and dc-link voltage are given by

$$\omega_{\text{gsc}} - \omega_0 = K_\theta (v_{\text{dc}} - v_{\text{dc}}^\star), \tag{14a}$$

$$P_{\text{wt}} - P_{\text{del}} = K_p^{\text{msc}} (v_{\text{dc}} - v_{\text{dc}}^\star). \tag{14b}$$

In other words, increasing K_θ results in larger steady-state dc voltage deviations relative to the system frequency deviation. Therefore, a general guideline for selecting K_θ is that the steady-state dc voltage should remain within acceptable limits (i.e., $\pm 5\%$) for an ac frequency within the nominal operating range (i.e., $\pm 1\%$), i.e., $0 < K_\theta \leq 0.2$ in per unit. The steady-state dc voltage limits used for control design are generally well within the hardware limits discussed in Sec. III-A to ensure sufficient margins for transients.

Moreover, combing these two equations results in

$$P_{\text{wt}} - P_{\text{del}} = K_p^{\text{msc}} K_\theta (\omega_{\text{gsc}} - \omega_0), \tag{15}$$

i.e., the wind turbine provides a steady-state primary frequency control response with effective frequency droop coefficient

$$m_p = \frac{1}{K_p^{\text{msc}} K_\theta}. \tag{16}$$

Given m_p (i.e., by grid-codes or markets), the corresponding proportional gain K_p^{msc} of the MSC dc voltage control gain can be calculated as $K_p^{\text{msc}} = 1/m_p 1/K_\theta$. In other words, decreasing the effective frequency droop coefficient m_p requires either increasing the MSC dc voltage control gain K_p^{msc} or increasing K_θ. In this work, the gains are selected by first increasing K_θ to its maximum value based on the steady-state map between GSC ac frequency deviation and dc voltage deviation, and then selecting according to $K_p^{\text{msc}} = 1/m_p K_\theta$.

A key feature of the controls (12) and (13) is that increased transient stability can be achieved through the derivative gains K_d^{gsc} and K_d^{msc}. In particular, increasing K_d^{gsc} results in faster synchronization through the ac network (i.e., decreased oscillations) and increased damping of the dc voltage. In particular, for systems with strong coupling and a GSC with low rating or large effective droop coefficient m_p, the gain K_d^{gsc} can be increased to ensure small-signal stability (see [45, Sec. V-A] for a discussion and analytical results). From a practical point of view, a large gain K_d^{gsc} may result in amplification of noise and higher order harmonics. In our case studies we observe that $0 < K_d^{\text{gsc}} \ll K_\theta$, i.e., a small non-zero value typically represents a good trade-off.

Finally, increasing K_d^{msc} results in increased damping of the dc voltage through the MSC at the expense of additional mechanical stress on the WT. The trade-off between these two aspects is difficult to quantify analytically and, for the purpose of our case study, simulations have been used to select a gain K_d^{msc} that represents a reasonable trade-off.

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F. Fault response and operational constraints

We emphasize that the controls proposed in Sec. IV-B and Sec. IV-C can be readily combined with existing heuristics for current limiting (e.g., threshold virtual impedance [46]) to achieve fault ride through capabilities and heuristic limiters to account for operational constraints (e.g., rotor speed and dc voltage limits). However, the choice of ac-GFM control typically does not have a significant impact on, e.g., the fault response. Instead, the choice of current limiting strategy largely determines the fault response [47]. Therefore, we chose to limit the scope of this work to normal grid conditions. Moreover, to the best of the authors' knowledge, no ac-GFM control design method exists that explicitly and systematically accounts for various constraints. Developing a more systematic framework for ac-GFM control of PMSG WTs subject to constraints and investigating the resulting fault response is seen as interesting topic for future work.

V. Case Studies and Comparisons

The test system shown in Fig. 10 is used to illustrate the main features of energy balancing GFM control and ac-GFM droop control under the proposed curtailment strategy. The system contains a synchronous generator and a wind farm. The 210 MVA synchronous generator operates at 0.8 power factor and is equipped with a turbine/governor systems and automatic voltage regulator. The droop coefficient of the governor is set to 5%. The wind farm consists of ten identical 5 MW PMSG-WTs with parameters shown in Table I. The active and reactive power drawn by the constant power load is 100 MW and 5 MVar and the load is increased by 20 MW during the simulation. Finally, we emphasize that frequency / active power and voltage / reactive power are commonly assumed to be decoupled for transmission connected plants during normal operation. Consequently, in this work, we focus on the active power response and resulting frequency dynamics. Moreover, all controls are tuned such that the converter current and dc-voltage limits will not be reached during normal operation to facilitate a fair comparison.

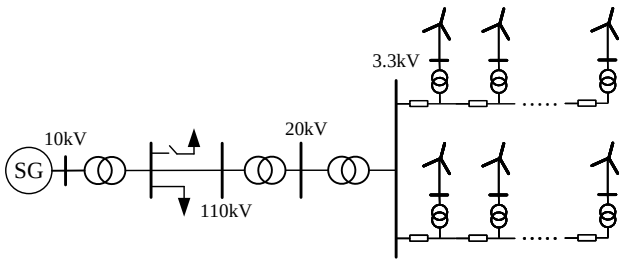


Fig. 10. Test system with 210 MVA synchronous generator, 100 MW/5 MVar constant power load, and 20 MW load that is initially disconnected.

A. Constant wind speed: below rated wind speed

We first assume that the wind speed is $v_{\mathrm{w}} = 10$ m/s throughout simulation and 20 MW load is connected at $t = 20$ s. To evaluate and compare the performance of different controls, the ac-GFL/dc-GFM PLL-based MPPT control (GFL+MPPT), energy balancing GFM control with MPPT control at the MSC (EB GFM+MPPT), energy balancing GFM control with grid supporting control at MSC (EB GFM+GS), ac-GFM/dc-GFL droop control with grid supporting MSC control and GSC power limit $P_{\max} = P_{\text{rated}}$ (Droop GFM+GS), and ac-GFM/dc-GFL droop control with grid supporting MSC control and GSC power limit $P_{\max} = P_{\text{mpp}}$ (Droop GFM+GS with limit), are implemented on the ten PMSG-WTs. Simulation results for all controls are shown in Fig. 11. We emphasize that the GSC frequency for ac-GFL control is an estimate obtained by a SRF-PLL and not directly comparable to the GSC frequencies shown for ac-GFM controls.

TABLE I
Wind turbine parameters

Component	Parameter	Value
WT	Rated power	5 MW
	Rated wind speed	11.23 m/s
	Total moment of inertia	$35.328 \times 10^6 \mathrm{kg} \cdot \mathrm{m}^2$
Converter	dc-link voltage	7.92 kV
	dc-link capacitor	79.711 mF
LCL filter	GSC filter resistance	0.0218 Ω
	GSC filter inductance	0.693 mF
	GSC filter capacitance	73 μF

The control parameters are selected based on the guidelines in Section IV-E and can be found in Table II. The focus of this work is active power control and the reactive power droop coefficient is set to a standard value. It can be seen that WTs using GFL+MPPT control do not respond to load disturbance and the dc-link voltage is maintained at 1 p.u. throughout simulation. Using this control, only the synchronous generator provides inertial response and primary frequency control. In this case, the frequency nadir and steady-state frequency after the load step are 49.39 Hz and 49.71 Hz (see Fig. 11h).

In contrast, using energy balancing GFM control the imbalance between load and generation is reflected in the dc voltage (see Fig. 11b). As a result, GSC frequency decrease when the load disturbance occurs. In other words, the GSC using energy balancing control behaves similarly to a synchronous generation (see Fig. 11c and Fig. 11h) and illustrates the GSC self-synchronization without PLL. Using MPPT control the MSC does not provide dc voltage regulation and therefore the PMSG-WT does not contribute to frequency regulation (see Fig. 11d) and the frequency nadir and steady-state frequency are identical to using PLL-based MPPT control. On the other hand, with 10% curtailment and grid-supporting MSC control, the WTs adjust their power generation to provide grid support. In this case, the frequency dynamics significantly improve with EB GFM+GS control and the frequency nadir and steady-state frequency increase to increases to 49.61 Hz and 49.75 Hz (see Fig. 11h). Because the MSC regulates the dc voltage, the dc voltage nadir and steady-steady state dc voltage are 0.984 p.u. and 0.99 p.u. compared to 0.976 p.u. and 0.988 p.u. for (EB GFM+MPPT). As shown in Fig. 11e and Fig. 11f, the kinetic energy stored in the WT is released after the load step and result in a temporary

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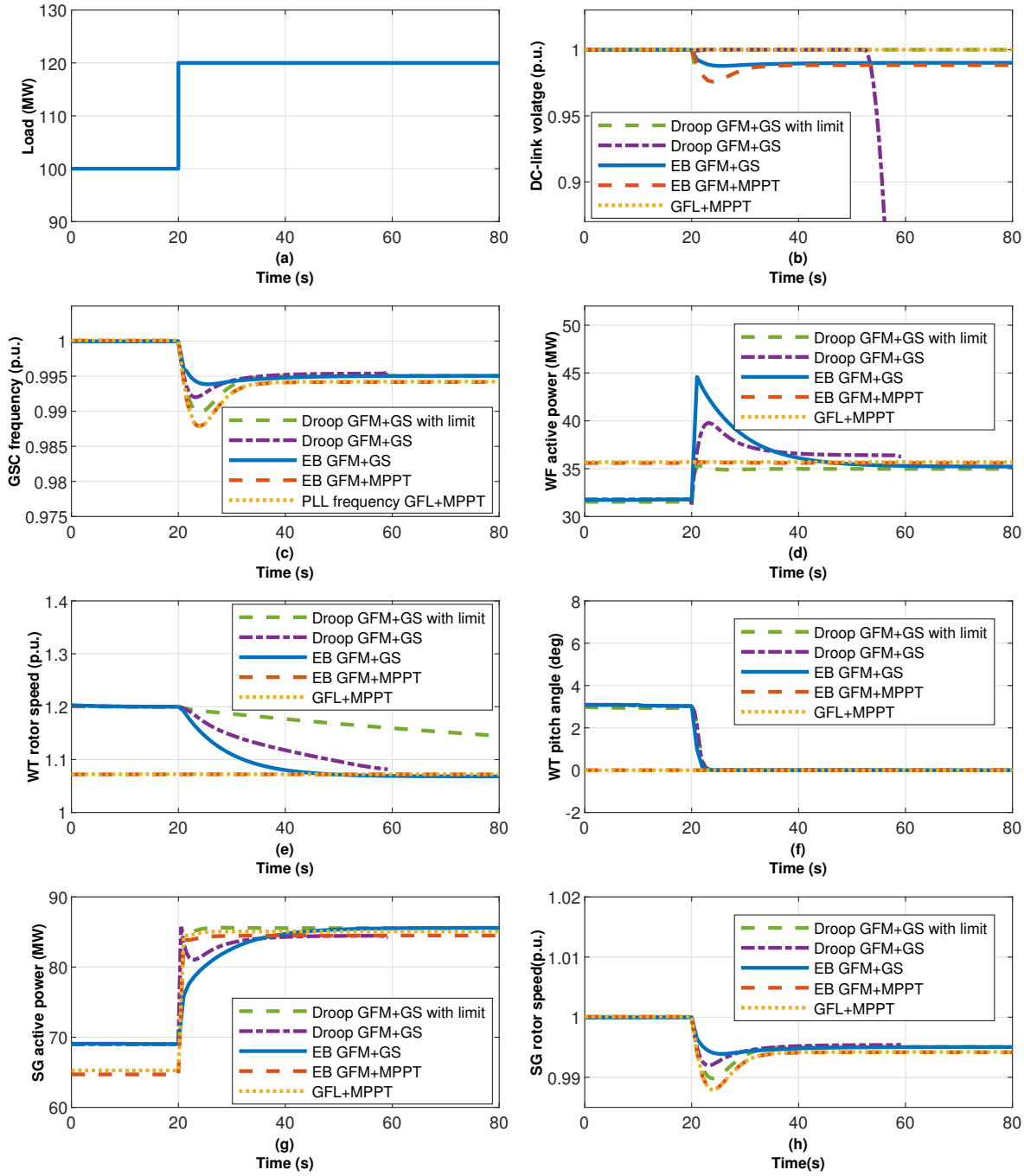


Fig. 11. Simulation results for $v_{\mathrm{w}} = 10$ m/s and ac-GFL/dc-GFM MPPT control (GFL+MPPT), energy balancing ac-GFM/dc-GFM control with MPPT MSC control (EB GFM+MPPT), energy balancing ac-GFM/dc-GFM control with MSC dc voltage control (EB GFM+GS), ac-GFM/dc-GFL droop control (Droop GFM+GS), and ac-GFM/dc-GFL droop control with GSC power limiting (i.e., $P_{\max} = P_{\mathrm{mpp}}$).

overload capability (see Fig. 11d) as the WTs converge to the MPP after the load disturbance.

On the other hand, using ac-GFL/dc-GFM droop control the system becomes unstable because the frequency-watt droop of the GSC overloads the WTs and the dc voltage collapses at approximately $t = 55$ s (see Fig. 11b). This is not surprising because the WTs cannot be regarded as ideal current sources, and without a proper coordinator between GSC controller and MSC controller, the WTs become unstable. Using energy balancing control this coordination is achieved through the dc voltage and the WTs remain stable. We emphasize that inertia emulation (i.e., increasing T_{ac}) would only exacerbate this issue further.

As discussed in Sec. IV-B, the instability observed for droop control may be resolved by setting the GSC power limit in ac-GFL/dc-GFM droop control to P_{mpp}. In this case, stable

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operation of the WT is maintained throughout simulation at the expense of not leveraging the kinetic energy storage of the WT and increased control complexity (i.e., the GSC control becomes nonlinear and tuning of controller is non-trivial).

Moreover, the time constant T_{ac} of ac-GFL/dc-GFM droop control can be used for inertia emulation (i.e., virtual synchronous machine control) with inertia constant $H^{\text{gsc}} = T_{\text{ac}}/(2K_p^{\text{ac}})$. The simulation results for frequency nadir, RoCoF, the maximum active power of the WT, and the minimum dc voltage are summarized in Table III and do not significantly differ from standard droop control (i.e., $T_{\text{ac}} = 0.02$ s) because GSC droop control power limiter precludes a significant inertia response that exceeds P_{mpp}.

In contrast, the EB GFM control explicitly controls the dc voltage through the GSC ac terminal if need be to ensure stability of the dc voltage when the GSC can no longer tightly control the dc voltage. Using this approach, the inherent kinetic energy storage of the WT is fully leveraged to provide inertia support.

On the other hand, both MPPT controls use the same MPPT control at the MSC. As a result, they do not provide grid-support and the rotor speed, pitch angle, and active power injection from the WT is the same with these two methods. The key difference between EB GFM+MPPT and GFL+MPPT lies in the control strategy at the GSC. The GFL+MPPT control implements a stiff proportional-integral (PI) control of the dc-link capacitor voltage through the GSC current. Consequently, the dc voltage does not respond to grid side disturbance. In contrast, EB GFM+MPPT uses a realizable PD controller to self-synchronize and control the dc-link capacitor voltage through the GSC ac terminal. Consequently, the dc voltage decreases after the load increase in the system, which in turn leads to the decrease of GSC frequency. This verifies that the WT can achieve self-synchronization without relying on a PLL.

B. Constant wind speed: low wind speed

Simulation results for a low wind speed scenario (i.e., $v_{\text{w}} = 6$ m/s) are shown in Fig. 13. In low wind speed conditions, curtailment using on rotor speed control is preferable over curtailment through pitch control to leverage the ability of the WT to store significant amounts of kinetic energy. As shown in Fig. 12 in the revised manuscript, MPPT control at a wind speed of 6 m/s results in a rotor speed of 0.643 p.u., pitch angle of 0 degree, and 0.154 p.u. active power injection. The results show that the proposed energy balancing grid forming control ensures the self-synchronization of WT both when implementing MPPT and grid support functions. The curtailment strategy results in a rotor speed of 0.872 p.u., i.e., the WT stores significant kinetic energy in the rotor. Once the load step occurs, the WT releases kinetic energy from the rotor and provides a significant inertia response. In this scenario, the primary control reserves are very limited due to the low wind speed. Therefore, the improvement to the steady-state frequency is not significant. Overall, under low wind speed condition, the amount of stored kinetic energy is significant while the primary reserve is limited. The key advantage of the proposed energy-balancing control in this operating regime is that it enables operation both at MPPT mode and grid supporting mode and transparently provides the appropriate amount of inertia and primary frequency response without requiring re-tuning of virtual inertia constants of droop coefficients based on the wind speed.

TABLE II
CONTROL PARAMETERS

K_θ	K_d^{gsc}	T_{dc}	K_q	T_{ac}
0.5 p.u.	0.025 p.u.	0.1 s	0.03 p.u.	0.02 s
K_p^{ac}	K_p^{msc}	K_d^{msc}	K_ω	
0.05 p.u.	40 p.u.	20 p.u.	1.5 p.u.	

TABLE III
RESPONSE OF AC-GFM/DC-GFL DROOP CONTROL WITH POWER LIMITING AND GRID-SUPPORT FUNCTIONS FOR DIFFERENT T_{AC}

T_{ac}	Frequency nadir	RoCoF	$P_{\max}$	$v_{\text{dc}}^{\min}$
0.02 s	49.490 Hz	0.127 Hz/s	35.314 MW	0.990 p.u.
0.4 s	49.490 Hz	0.129 Hz/s	35.263 MW	0.987 p.u.
0.8 s	49.490 Hz	0.129 Hz/s	35.257 MW	0.987 p.u.

C. Constant wind speed: high wind speed

Simulation results for a high wind speed scenario (i.e., $v_{\text{w}} = 12$ m/s) are shown in Fig. 12. In the high wind speed scenario WT rotor speed limits preclude increasing the rotor speed ω_r beyond $\omega_{r,\text{opt}}$ and curtailment can only be achieved through pitch angle control. Similarly to the scenario with $v_{\text{w}} = 10$ m/s, the WT loses stability when using ac-GFM droop control with $P_{\max} = P_{\text{rated}}$. Stable operation of WT can be maintained via ac-GFM droop control with $P_{\max} = P_{\text{mpp}}$ and using energy balancing GFM control. Notably, a significant inertia response is provided by EB GFM control in the low wind speed scenario. In contrast, in the high wind speed scenario, we can (i) no longer curtail the wind power generation through increasing the rotor speed, and (ii) no inertia response can be provided because the rotor speed is already at the optimal rotor speed $\omega_{r,\text{opt}}$ and no kinetic energy can be released from the rotor without decreasing wind power generation. Instead, pitch angle control is used for curtailment and to provide a fast frequency response. Broadly speaking, the proposed control still results in a fast frequency response. However, the active power response of the wind turbine is slowed down by the pitch motor time constant. These results illustrate that the EB GFM control automatically provides an appropriate inertia response without requiring re-tuning of a virtual inertia constant depending on the wind speed.

The frequency nadir, RoCoF, maximum active power injection of the WT, and the minimum dc voltage for different wind speeds and control strategies are summarized in Table IV. Overall, the results verify that the EB GFM outperforms the other strategies with respect to standard performance metrics.

D. Constant wind speed: islanding of the WTs

We also test the scenario that the system switches to island mode at 30 s (see Fig. 14). A wind speed of 10.5 m/s is

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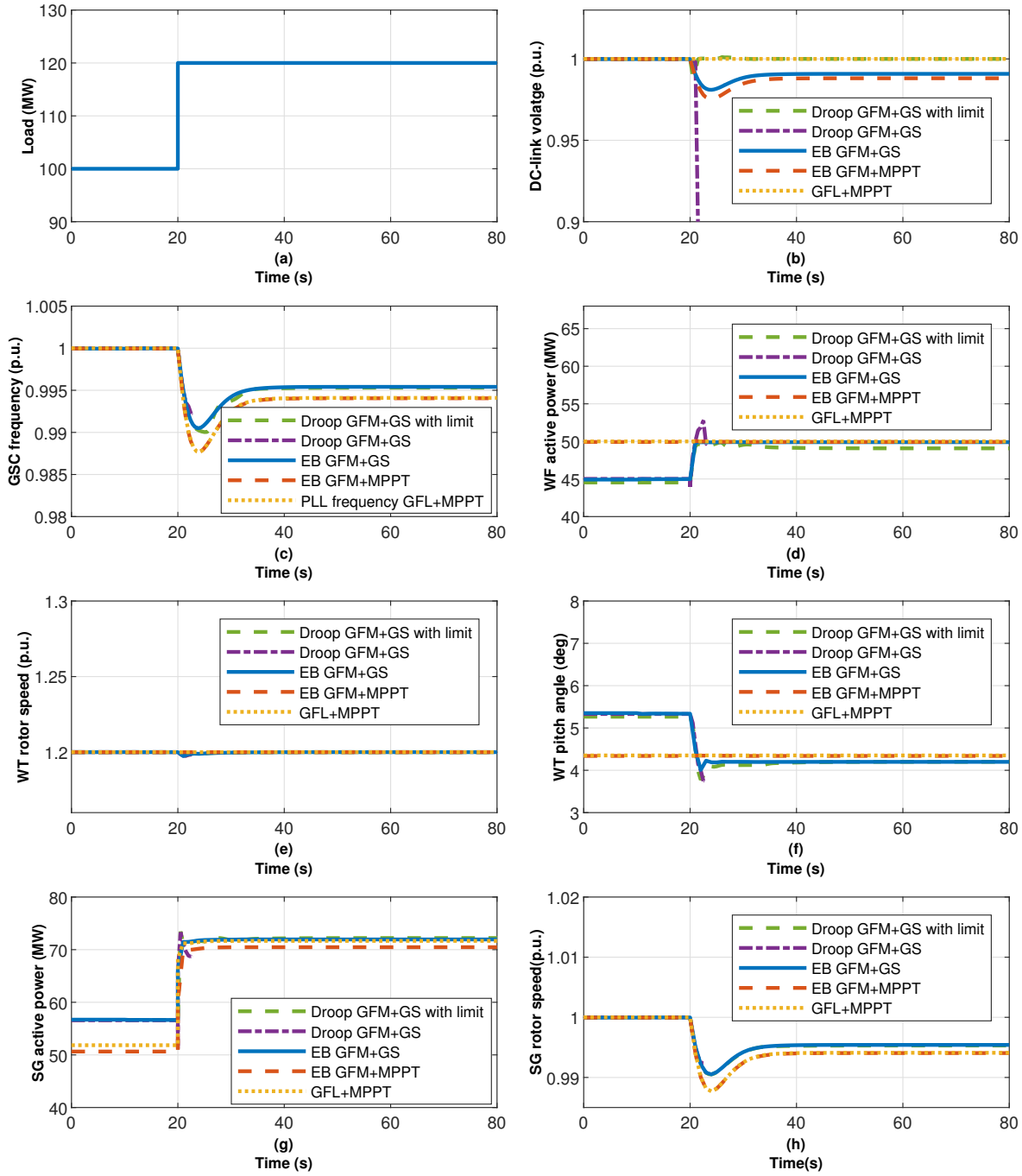


Fig. 12. Simulation results for $v_{\mathrm{w}} = 12$ m/s and ac-GFL/dc-GFM MPPT control (GFL+MPPT), energy balancing ac-GFM/dc-GFM control with MPPT MSC control (EB GFM+MPPT), energy balancing ac-GFM/dc-GFM control with MSC dc voltage control (EB GFM+GS), ac-GFM/dc-GFL droop control with MSC dc voltage control (Droop GFM+GS), and ac-GFM/dc-GFL droop control with GSC power limiting (i.e., $P_{\max} = P_{\mathrm{mpp}}$).

used throughout simulation, and the constant power load is 30 MW in island mode. Since the ac-GFL control relies on the synchronous generator to form a stable PCC voltage, the system collapses after switching to island mode. In contrast, with the proposed EB GFM control, the WT forms a stable voltage at GSC terminal and supports the stable operation of the island system. As the load demand in the island mode is less than wind power generation in grid-connected mode, the GSC frequency has a slight increase at 30 ss. As a result, the dc-link voltage increases once the system is islanded.

E. Time-varying wind speed

Next, PLL-based ac-GFL/dc-GFM MPPT control, droop ac-GFM/dc-GFL control, and energy-balancing GFM control with and without grid support functions are compared in a simulation using the time-varying wind speed profile shown in

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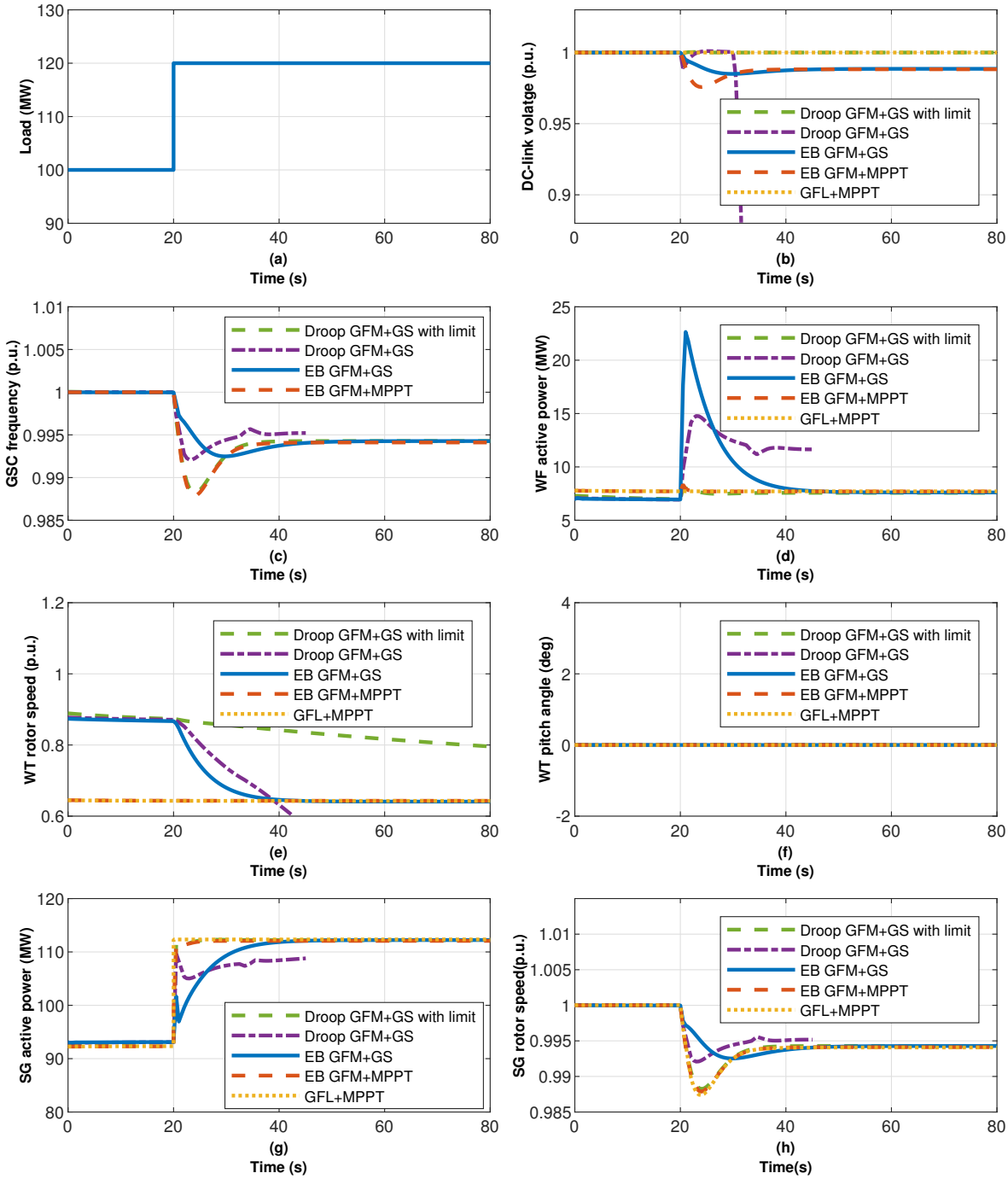


Fig. 13. Simulation results for $v_{\mathrm{w}} = 6$ m/s and ac-GFL/dc-GFM MPPT control (GFL+MPPT), energy balancing ac-GFM/dc-GFM control with MPPT MSC control (EB GFM+MPPT), energy balancing ac-GFM/dc-GFM control with MSC dc voltage control (EB GFM+GS), ac-GFM/dc-GFL droop control with MSC dc voltage control (Droop GFM+GS), and ac-GFM/dc-GFL droop control with GSC power limiting (i.e., $P_{\max} = P_{\mathrm{mpp}}$).

Fig. 15a. Moreover, the 20 MW load is connected at $t = 150$ s. The simulation results are shown in Fig. 15b to Fig. 15h.

Again, the WTs with MPPT control do not respond to imbalances between load and generation and GFL+MPPT control controls the dc voltage to 1 p.u. throughout the simulation (see Fig. 15b and Fig. 15d). Using ac-GFM/dc-GFL $P-\omega$ droop control, the dc-link voltage collapses at about 42 s because the wind turbine cannot provide the required active power due to its operation limits. Using energy-balancing GFM control, the dc voltage again reflects power imbalances (see Fig. 15b) and GSC frequency self-synchronizes with the synchronous generator (see Fig. 15b and Fig. 15h). The combination of fluctuating wind power generation and the load disturbance result in frequency deviations shown in Fig. 15h. The standard deviation of the frequency across the simulation is 0.223 Hz with GFL+MPPT and EB GFM+MPPT

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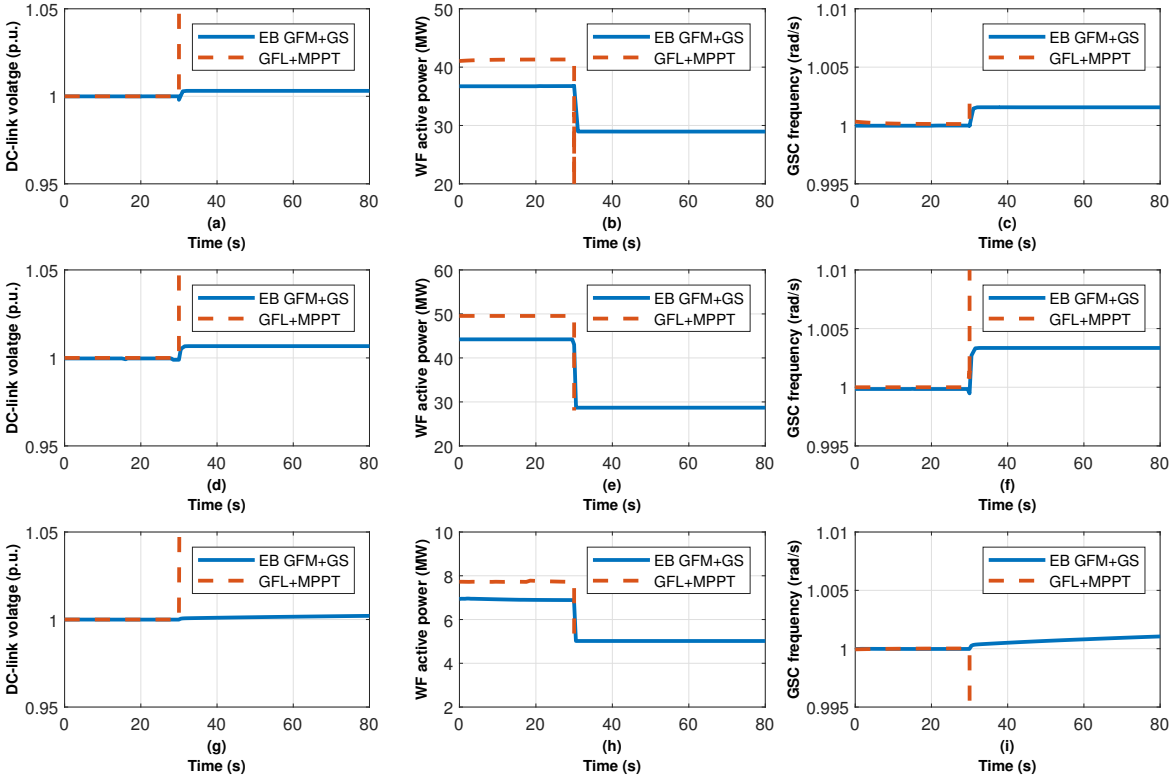


Fig. 14. Islanding of the WT for $v_{\rm w} = 10.5$ m/s, $v_{\rm w} = 12$ m/s, $v_{\rm w} = 6$ m/s, PLL-based MPPT control (GFL+MPPT), and energy balancing ac-GFM/dc-GFM control (EB GFM+GS).

TABLE IV
COMPARISON OF DIFFERENT CONTROL STRATEGIES

v_w	Control strategy	Frequency nadir (Hz)	RoCoF (Hz/s)	$P_{\max}$ (MW)	$v_{\rm dc}^{\min}$ (p.u.)
6 m/s	GFL +MPPT	49.393	0.167	7.703	1.000
	EB GFM+MPPT	49.393	0.167	7.703	0.976
	Droop GFM +GS with limit	49.446	0.150	8.133	0.994
	EB GFM+GS	49.628	0.060	22.969	0.985
8 m/s	GFL +MPPT	49.392	0.167	18.250	1.000
	EB GFM+MPPT	49.392	0.167	18.250	0.976
	Droop GFM +GS with limit	49.415	0.140	18.371	0.991
	EB GFM+GS	49.685	0.027	32.879	0.987
10 m/s	GFL +MPPT	49.393	0.167	35.600	1.000
	EB GFM+MPPT	49.393	0.167	35.600	0.976
	Droop GFM +GS with limit	49.490	0.129	35.263	0.987
	EB GFM+GS	49.680	0.081	44.249	0.987
12 m/s	GFL +MPPT	49.384	0.167	49.927	1.000
	EB GFM+MPPT	49.384	0.167	49.927	0.976
	Droop GFM +GS with limit	49.524	0.117	49.641	0.990
	EB GFM+GS	49.515	0.118	49.398	0.981
14 m/s	GFL +MPPT	49.384	0.167	49.927	1.000
	EB GFM+MPPT	49.384	0.167	49.927	0.976
	Droop GFM +GS with limit	49.524	0.117	49.641	0.990
	EB GFM+GS	49.515	0.118	49.398	0.981

control. With energy-balancing GSC control, MSC dc voltage control, and 10% curtailment, the WTs contribute to frequency regulation and frequency deviations induced by both time-varying wind speeds and the load disturbance are effectively reduced. In this case, the standard deviation of the frequency across the simulation decreases to 0.175 Hz (i.e., decreases by 21.5% compared to MPPT control). As a result, the frequency regulation burden on the synchronous generator is mitigated (see Fig. 15g). As shown in Fig. 15e and Fig. 15f the WT rotor speed and pitch angle remain within acceptable limits throughout the simulation.

Finally, we investigate the impact of the controls on mechanical stress and fatigue loads of the WTs [48]. In particular, the shaft torque and thrust force are given by

$$T_{\rm sh} = \frac{P_{\rm wt}}{\omega_{\rm r}} - J\frac{\rm d}{{\rm d}t}\omega_{\rm r}, \tag{17a}$$

$$F_{\rm t} = \frac{1}{2}\rho\pi R^2 C_t(\lambda,\beta)v_{\rm w}^2, \tag{17b}$$

and the thrust coefficient $C_t(\lambda,\beta)$ that depends on the tip speed ratio λ and pitch angle β. As shown in Fig. 15(e) and (f), the rotor speed and pitch angle significantly change when grid support is provided. To quantify the impact of different control strategies on fatigue loads on the WT shaft and WT tower, the so-called damage equivalent load (DEL) is calculated using MCrunch [49]. The resulting damage equivalent loads are listed in Table V. Compared to MPPT control using either a PLL or energy-balancing, using energy-balancing control or droop control with grid support, the shaft

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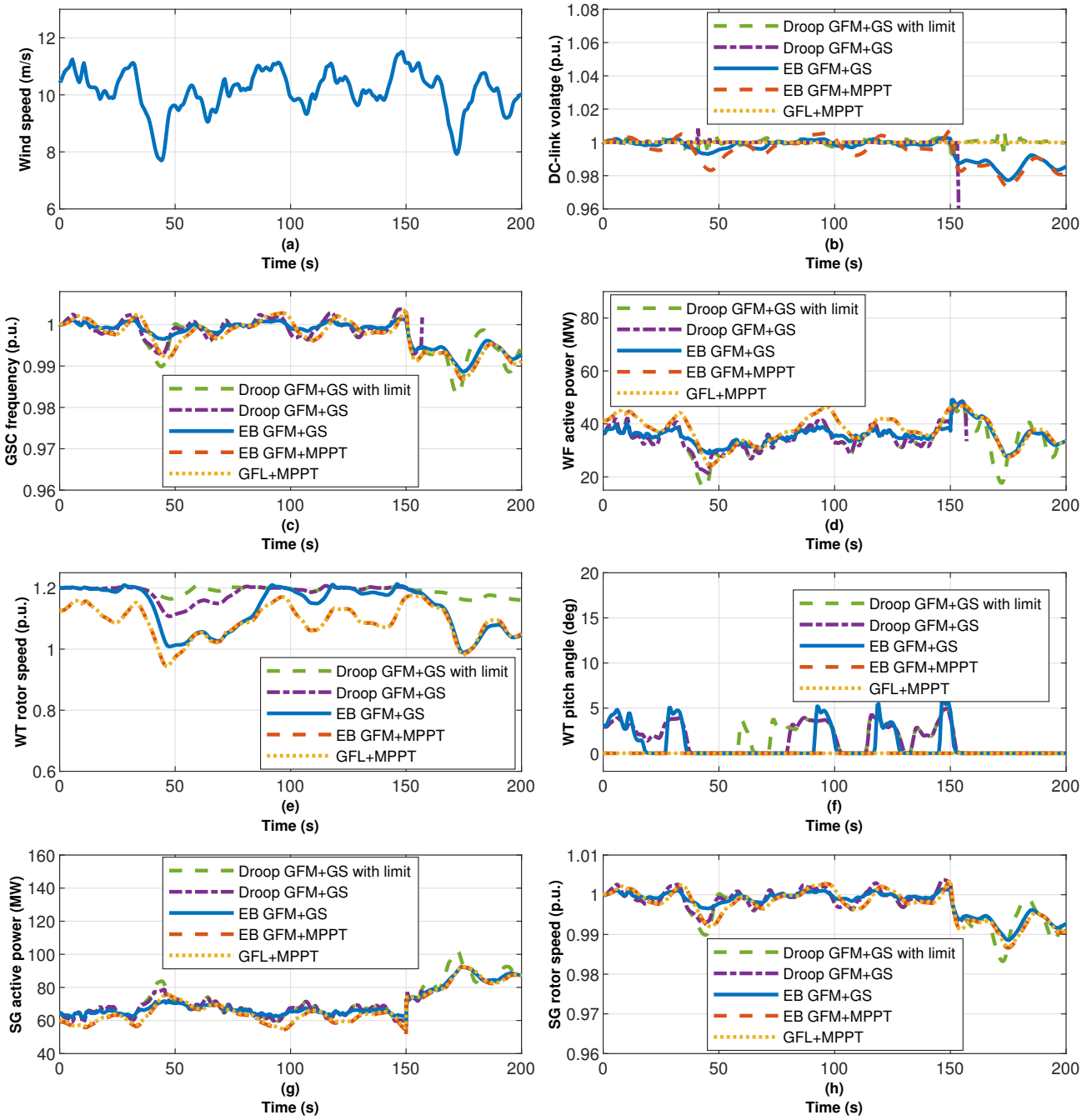


Fig. 15. Simulation results for the time-varying wind speed and ac-GFL MPPT control (GFL+MPPT), energy balancing ac-GFM/dc-GFM control with MPPT MSC control (EB GFM+MPPT), energy balancing ac-GFM/dc-GFM control with MSC dc voltage control (EB GFM+GS), ac-GFM/dc-GFL droop control with MSC dc voltage control (Droop GFM+GS), and ac-GFM/dc-GFL droop control with GSC power limiting (i.e., $P_{\max} = P_{\text{mpp}}$).

damage equivalent load has a slight increase compared to MPPT control, while the tower damage equivalent load has a slight decrease. The energy captured by the WTs throughout the simulation also decreases by 5.37% when using energy-balancing control with frequency regulation (2095.717 kWh with MPPT control and 1982.122 kWh with frequency regulation). We emphasize that in power systems with large-scale renewable integration, economic objectives such as maximizing wind power production have to be balanced with ensuring stable system operation. From this viewpoint optimizing the trade-off between wind power generation, fatigue loads, and frequency support is seen as an interesting topic for future work.

VI. Conclusion

This work proposes a grid-forming control architecture for PMSG WTs that supports the entire spectrum from MPPT to grid-support functions (i.e., inertia response and

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TABLE V
WT FATIGUE LOADS

Control	Shaft DEL	Tower DEL
MPPT (GFL & EB)	1.991 MNm	39.928 MNm
EB GFM+GS	2.001 MNm	38.645 MNm
Droop GFM + GS	1.955 MNm	37.204 MNm

primary frequency control) without control mode switching. The proposed control architecture combines energy-balancing dual-port GFM control of the GSC with a suitable curtailment and MSC/WT control scheme to fully leverages the WT capabilities to provide GFM grid-support. Comprehensive case studies (constant and time-varying wind speed scenarios, grid-connected and islanded scenarios) have been used to evaluate and compare the dynamic performance of the proposed control to state-of-the-art controls. The results demonstrate that the proposed energy-balancing control for PMSG-WTs outperforms standard PLL-based ac-GFL/dc-GFM and ac-GFM/dc-GFL controls with respect to standard metrics. The proposed control provides a promising pathway to future power systems with high penetration of converter based resources. Extending the proposed controls to systematically account for constraints (e.g., current limits) and investigating the fault ride through capabilities of PMSG-WTs with grid-forming controls is seen as interesting topic for future research. Moreover, future work will extend the stability analysis framework developed in [24] to include the proposed energy-balancing grid-forming control.

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Xue Lyu (S'16 - M'19) received her Ph.D. degree in electrical engineering from The Hong Kong Polytechnic University, Hong Kong, in 2019. She is currently a staff research engineer at Pacific Northwest National Laboratory. Prior to joining PNNL, she was a research associate at University of Wisconsin-Madison from 2021 to 2022 and a postdoctoral fellow at The University of Hong Kong from 2019 to 2020. Her research interests include control and optimization for renewable energy integration in power systems.



Dominic Groß (S'11 - M'15) is an Assistant Professor with the Department of Electrical and Computer Engineering at the University of Wisconsin-Madison, Madison, WI, USA. He received his Ph.D. degree in Electrical Engineering from the University of Kassel, Germany, in 2014. Prior to joining UW-Madison, he was a postdoctoral researcher at the Automatic Control Laboratory of ETH Zürich. He received an NSF CAREER award in 2022. His research interests include distributed control and optimization of complex networked systems with applications in converter-dominated power systems and grid-forming control of power electronics.

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